Multimodal Remote Sensing Object Detection Based on Prior-Enhanced Mixture-of-Experts Fusion Network

Kewei Liu, Dongliang Peng, and Tao Li

***Abstract*— Multimodal remote sensing image object detection enhances detection accuracy by fusing complementary information from multimodal image data. However, complex environments significantly affect the reliability and complementarity of multimodal data, and traditional methods struggle to dynamically adapt to environmental changes, leading to degraded detection performance. To address this challenge, in this article, we propose a multimodal remote sensing object detection method based on a prior information-enhanced mixture-of-experts fusion network (PMFN). Specifically, we first introduce a PMFN framework to achieve environment-adaptive multimodal image feature fusion. Second, we propose a dynamic gating network that combines prior information and multimodal image features to endow the system with environmental perception capabilities. This network is employed to dynamically allocate weights to subfusion experts optimized for different environmental conditions within the mixture-of-experts (MoE) fusion network framework. Furthermore, to fully exploit the complementary information present in multimodal image features, we propose a frequency-decoupled feature fusion network as a subfusion expert within the MoE fusion network framework. This utilizes wavelet transform to decouple the features of each modality and then develops personalized fusion strategies for each frequency subband. In addition, to enhance detection efficiency, we introduce a cross-scale feature channel interleaved fusion (CSFCIF) strategy, which significantly reduces computational cost while ensuring stable detection performance. Experimental results on the DroneVehicle and RGBT-Tiny datasets demonstrate that our method achieves competitive performance compared to state-of-the-art approaches. Code will be available at: https://github.com/LiuKewei0110/MDPMFN**

***Index Terms*— Multimodal object detection, multispectral fusion, multispectral object detection, remote sensing object detection, RGB–infrared object detection.**

I. Introduction

OBJECT detection is a fundamental task in remote sensing, playing a key role in applications such as smart transportation, urban security, and disaster rescue. However, relying on a single-sensor modality (e.g., visible or infrared) often proves insufficient due to complex environmental interference and varying lighting conditions [1].

The performance of object detection using only visible or infrared images is limited by factors like weather conditions and the intrinsic characteristics of the imaging systems [2]. Visible images, for instance, capture rich textures and fine details under adequate illumination, while infrared images excel in revealing temperature differences in low-light environments [3].

Recognizing the complementary strengths of these modalities, the fusion of visible and infrared remote sensing images has emerged as an effective approach to enhance ground object detection [4], [5]. Multimodal and cross-sensor approaches have also shown promise in diverse sensing conditions [6], [7]. This multimodal fusion strategy fully leverages the strengths of images from different modalities, thereby enhancing accuracy in challenging conditions.

In recent years, various multimodal modeling techniques have been explored across different visual tasks, further demonstrating the potential of cross-modal representation learning in challenging environments [8], [9]. Visible images are acquired using visible sensors based on CMOS technology. Due to their low cost and ease of acquisition, such sensors are widely deployed on various aerial and spaceborne platforms. As a result, the acquisition of remote sensing visible images has become increasingly convenient, providing strong support for the rapid advancement of remote sensing object detection and significantly driving technological progress and expanding its applications. This has led to the emergence of numerous advanced detection algorithms, including two stage methods, such as Faster R-CNN [10], RoITransformer [11], Gliding Vertex [12], and ReDet [13], as well as single stage methods, such as RetinaNet [14], S2ANet [15], and R3Det [16]. These algorithms have greatly enhanced the capability of remote sensing image interpretation, further propelling the development of the field. Under low-light and poor visibility conditions, the performance of remote sensing object detection algorithms based on visible images tends to degrade drastically. In such environments, infrared imagery offers a distinct advantage. Since infrared images are unaffected by illumination variations, their detection performance remains relatively stable despite changes in lighting conditions. Numerous outstanding studies have been conducted in this field, including MSHNET [17], ACM [18], DNAnet [19], and ALCNet [20].

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Although infrared images are unaffected by illumination variations, their detection performance still lags behind that of visible images under well-lit conditions. As a result, increasing attention has been given to the fusion of infrared and visible images for object detection. The release of datasets, such as DroneVehicle [5] and RGBT-Tiny [21], has laid an important foundation for advancements in this field, while works like UA-CMDet [5] and calibrated and complementary transformer (C2Former) [4] have further accelerated its development.

Current infrared and visible image fusion-based object detection methods suffer from limited robustness in complex environments due to their uniform fusion strategies that fail to account for the impact of the environment on the feature quality of visible and infrared images. Environmental variations can induce pronounced disparities in the quality of infrared and visible images, thereby engendering significant differences in the complementary nature of the information they convey. For instance, under favorable illumination and visibility conditions, infrared and visible images exhibit strong similarity in low-frequency features, whereas this similarity diminishes significantly in poorly lit or low-visibility scenarios. Such uncertainty in the reliability and relevance of multimodal features caused by environmental changes poses challenges for enhancing the performance of detection algorithms. The grayscale histogram intuitively represents the brightness distribution and contrast information of an image, while information entropy effectively reflects the richness of information contained within the image. Utilizing the grayscale histogram and information entropy as prior information provides crucial information for assessing the illumination conditions and visibility status of the current environment. For instance, if the grayscale histogram of the visible image shifts toward the low-intensity range, and its information entropy is significantly lower than that of the infrared image, this indicates that the environment is likely in a low-light state. When the grayscale histogram of the visible image is concentrated in the medium-to-high intensity region, and its information entropy is notably lower than that of the infrared image, it often implies that the environment may be shrouded in haze or smoke. Conversely, if the grayscale histogram of the visible image is centered in the medium-intensity range, and the difference in information entropy between the visible and infrared images is minimal, it generally suggests that the environment has good lighting conditions and visibility (Fig. 1). However, existing visible–infrared fusion detection frameworks have yet to fully utilize critical prior information related to visible and infrared images, especially regarding illumination conditions and visibility levels.

To address these limitations, we propose a multimodal image fusion detection method based on a prior information-enhanced mixture-of-experts fusion network (MDPMFN). In this approach, a mixture-of-experts (MoE) fusion network is designed that comprises a prior information-assisted dynamic gating network (PGN) and a set of subfusion experts based on frequency-decoupled feature fusion, thereby achieving environment-adaptive multimodal feature fusion. The dynamic gating network, equipped with environmental perception and multimodal feature quality assessment capabilities, is responsible for the dynamic allocation of weights among the subfusion experts. The subfusion experts perform frequency decoupling on the multimodal features and are tasked with carrying out personalized fusion for different frequency subbands. In addition, each subfusion expert is individually optimized for various environmental and feature conditions. Moreover, to improve computational efficiency, we propose a cross-scale feature interleaved fusion strategy for the cross-scale feature interaction process, which effectively reduces the computational load while maintaining stable detection performance.

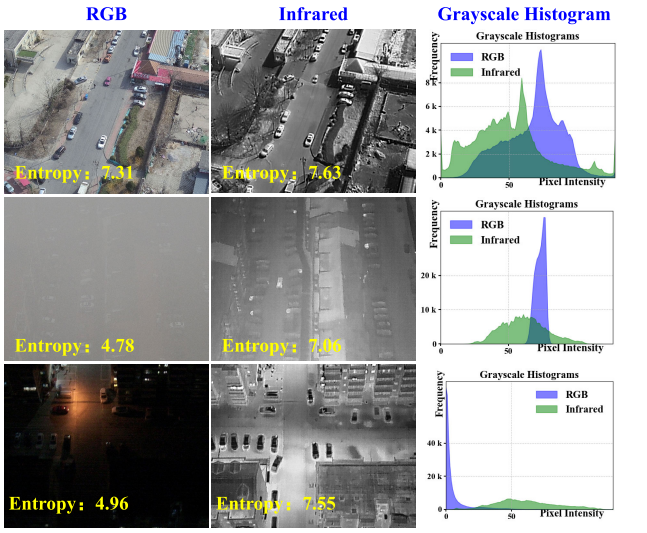


Fig. 1. Information entropy values and grayscale histograms of RGB and infrared images under different environmental conditions. The first row represents daytime conditions, the second row represents foggy conditions, and the third row represents low-light conditions. In the grayscale histogram, the blue color denotes the grayscale distribution of the visible image, while the green color represents that of the infrared image.

Existing methods, such as C2Former and UA-CMDet, generally adopt fixed or rule-based fusion strategies in multimodal integration, where the modality contributions remain predetermined or manually adjusted, regardless of varying environmental conditions. For instance, UA-CMDet heuristically adjusts fusion weights based on grayscale statistics and intermodal misalignment using handcrafted thresholds, which limits its adaptability in complex scenes involving dynamic illumination, cloud occlusion, or cluttered backgrounds. In contrast, this article proposes a fusion network based on a MoE architecture. Each subfusion expert within the MoE is optimized for specific environments, and a gating network driven by prior information dynamically adjusts the fusion strategy according to environmental conditions. Compared with fixed-weight fusion approaches, this design significantly enhances the model's generalization ability. Furthermore, the subfusion experts employ a frequency decoupling-based fusion strategy. Traditional frequency decoupling methods in image fusion generally ignore the differences between high- and low-frequency subband characteristics. This article proposes tailored fusion strategies for each subband, effectively preserving their distinct frequency-specific properties. Finally, in this article, prior information are used as one of the inputs to the gating network, enabling it to

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learnably adjust gating weights for implicit environmental partitioning, thereby replacing the explicit threshold-based heuristics used in previous methods. Instead of relying on hard-coded environmental classification or separate condition estimation branches, we extract prior cues directly from the image pair. These cues are fed into the gating network to inform expert selection, allowing the model to adapt its fusion behavior without requiring manual labels or rigid fusion rules. This data-driven perception mechanism equips the framework with superior flexibility in handling diverse and dynamically changing remote sensing conditions.

In summary, the contributions of this article are as follows.

1) We propose a prior information-enhanced mixture-of-experts fusion network (PMFN) composed of a PGN and a set of subfusion experts based on frequency decoupling. This network leverages a gating mechanism to precisely regulate collaboration among the subfusion experts, achieving environment-adaptive multimodal feature fusion and effectively enhancing the robustness of the detection method in complex environments.
2) We introduce a PGN. By integrating prior information such as grayscale histograms and information entropy of multimodal images with multimodal features, this network endows the system with environmental perception capabilities. It enables adaptive weight allocation to subfusion experts optimized for different environments within the MoE fusion network framework based on environmental conditions.
3) We propose a subfusion expert network based on frequency decoupling of multimodal features. This network decomposes multimodal image features into different frequency subbands and formulates personalized fusion strategies for each subband, enabling complementary integration of multimodal feature advantages.
4) We propose a cross-scale feature interleaved fusion strategy. This strategy aligns cross-scale features using upsampling or downsampling operations, then interleaves channels corresponding to the same relative level and performs dimensionality reduction-based fusion. It effectively reduces computational complexity while maintaining stable detection performance.

II. Related Work

A. Multimodal Remote Sensing Object Detection

Object detection in infrared and visible remote sensing images refers to the detection of ground objects by fusing infrared and visible images. The integration of such multimodal data utilizes the complementary information of the two imaging modalities, which can significantly improve the robustness and applicability of object detection. Much research in this field focuses on optimizing multimodal images at the feature level. Sun et al. [5] proposed a visible–infrared detection framework (UA-CMDet) based on uncertainty learning, which uses cross-modal intersection-over-union (IoU) and illumination estimation to evaluate the uncertainty weights of objects, thereby suppressing the bias of high-uncertainty objects during model training and enhancing the fusion effect in the inference stage through illumination-aware nonmaximum suppression.

In terms of handling modality differences, Chen et al. [22] proposed an offset-guided adaptive feature alignment (OAFA) method, which combines the cross-modality spatial offset modeling (CSOM) module and the offset-guided deformable alignment and fusion (ODAF) module to effectively address the weak alignment problem. Transformers have become a focus of attention due to their powerful feature correlation modeling capabilities. Yuan and Wei [4] proposed a C2Former, which calibrates features and enhances intermodal complementarity through a cross-modal attention mechanism; they also proposed the CAGT method, which aligns features using the translation–scale–rotation alignment (TSRA) module and captures complementary features with the complementary fusion transformer (CFT) module, and the constructed CAGTDet detector performs excellently in experiments.

Furthermore, He et al. [23] designed a cross-modal conflict-aware learning network (CALNet), which can effectively alleviate the heterogeneity problem and optimize the semantic fusion effect. To make the fusion mechanism more efficient, Qingyun and Zhaokui [1] developed a cross-modality attentive feature fusion (CMAFF) module, which improves the adaptability and flexibility of feature fusion by separately inferring attention maps for common and differential modalities; while Zhao et al. [24] designed the redundant spectrum removal and dynamic feature selection modules (RSDet) from coarse to fine, achieving efficient removal of interfering features and effective fusion. The specificity-guided cross-modal feature reconstruction (SCFR) algorithm proposed by Sun et al. [25] significantly improves the detection performance by focusing on the unique correlations between modalities.

Meanwhile, Sun et al. [26] proposed the LF-MDet method based on low-rank frequency expert encoding, which achieves high detection accuracy while reducing computational complexity. An et al. [27] designed a effectiveness guided cross-modal information sharing network (ECISNet) that ensures robust detection capabilities in single -modal failure scenarios by introducing the modal effectiveness guiding (MEG) module. And Wang et al. [28] further enhanced the expression quality and anti-interference ability of fusion features through joint representation optimization and specific information enhancement strategies.

Current research encompasses various methods from uncertainty modeling, cross-modal attention mechanisms, to weak registration optimization, providing rich technical means to solve the problem of infrared and visible image fusion detection. However, the significant differences in image quality in different environments and the insufficient adaptability of a single-fusion strategy to complex scenes remain difficult problems to be solved. Therefore, this article designs the MDPMFN method to meet the requirements of efficient and robust object detection in diverse scenarios.

B. Mixture-of-Experts

The MoE model is a flexible and powerful architecture that has demonstrated strong adaptability and performance

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in complex perception tasks. In recent years, researchers have explored and optimized MoE techniques from various perspectives, leading to valuable advancements across multiple domains [29]. Liu et al. [30] proposed the Gaussian process-gated hierarchical MoEs (GPHMEs) model, which employs Gaussian processes (GPs) to construct gating functions and expert networks, enabling nonlinear partitioning and efficient data processing. Compared with traditional tree-structured hierarchical MoE models, GPHMEs achieve certain performance improvements while offering better interpretability for deep GPs and Bayesian neural networks. In the field of industrial soft sensors, Huang et al. [31] introduced the balanced MoE (BMoE) model, which incorporates multigate MoE modules and task-gradient balancing mechanisms to optimize task relationships, aiming to enhance data utilization efficiency in multitask learning and mitigate negative transfer effects. In the field of image fusion, Cao et al. [32] proposed a multimodal gated local-to-global expert fusion framework (MoE-Fusion), which dynamically extracts effective infrared and visible image information through local expert networks (MoLE) and global expert networks (MoGE) to optimize cross-modal texture and contrast. For visual saliency prediction, Dodge and Karam [33] introduced MxSalNet, which integrates a semantics-guided gating network within an MoE framework to aggregate predictions from different expert networks, thereby achieving certain performance improvements.

Although existing MoE methods have made positive progress in multimodal perception and complex task modeling, there are still areas that warrant further investigation. For instance, there is considerable room for improvement in knowledge utilization, as well as in reducing redundancy and mitigating repetitive learning among experts.

To address these issues, this article proposes a prior knowledge-guided MoE network architecture. Within this framework, a prior-assisted dynamic gating mechanism is proposed, along with a sparsity–diversity constraint strategy among experts. These mechanisms are designed to optimize expert collaboration and enhance the model's generalization ability.

C. Wavelet Transform

Frequency features enhance model robustness by capturing multiscale and directional information. For instance, Zheng et al. [34] proposed a frequency-domain orientation learning module combining a frequency feature extraction network with an enhanced self-attention layer, significantly improving oriented object detection. Wang et al. [35] utilized frequency-domain disentanglement with learnable filters to extract domain-invariant and domain-specific spectra, enhancing generalization in complex scenarios. Similarly, Fu et al. [36] introduced a spatial–spectral feature pyramid to adapt global dependencies, enabling precise detection of oriented objects. Liu and Jiang [37] emphasized the complementary fusion of multifrequency features from different modalities, achieving robust performance in challenging scenes, while Tang et al. [38] used wavelet-based frequency disentanglement to refine change detection tasks and enhance robustness. In addition, wavelet-based convolutional layers, like WTConv [39], achieve large receptive fields with fewer parameters, benefiting multifrequency modeling.

However, these methods lack detailed analysis of frequency subband characteristics. In contrast, our approach analyzes subband features of remote sensing infrared and visible images and designs tailored fusion strategies for each, effectively integrating complementary modal information and strengthening the basis for object detection.

III. Proposed Method

The framework of the proposed algorithm is shown in Fig. 2. Infrared and visible images are fed into a Siamese backbone network to extract multiscale features for both modalities. Subsequently, a PMFN is employed to perform environment-adaptive fusion of the features from infrared and visible images. Within PMFN, each subfusion expert fuses the infrared and visible features based on its respective strengths. At the same time, the PGN in PMFN integrates the prior information with the infrared and visible image features to achieve environmental perception, thereby adaptively assigning weights to the subfusion experts optimized for different environments. Next, in the neck network of the object detection method, a cross-scale feature channel interleaving fusion module is used to facilitate the interaction of multiscale feature information. Finally, the detection results for remote sensing targets are obtained based on the processed features.

A. Prior Information-Enhanced MoE Fusion Network

Due to the limitations of imaging sensors, the information reliability of multimodal images and the complementarity between modalities exhibit significant variations under different environmental conditions. For instance, in well-lit and high-visibility environments, visible images can capture abundant and high-quality information. However, under adverse conditions such as low illumination or heavy fog, visible images may fail to provide effective information. Consequently, although a single-fusion model can achieve balanced performance in most scenarios, it may not deliver optimal results in specific conditions.

In contrast, the proposed PMFN offers distinct advantages. It integrates multiple subfusion expert modules designed specifically for the fusion of features from infrared and visible images. These subfusion expert modules can selectively fuse and optimize image information based on varying environmental characteristics and modality-specific properties, thereby significantly enhancing the model's adaptability and accuracy across diverse complex scenarios (Fig. 3) .

Specifically, the PMFN consists of a dynamic gating network that incorporates prior information and a set of subfusion experts based on frequency decoupling. The dynamic gating network analyzes the prior information along with infrared and visible features to perceive the current environment, assigning adaptive weights to subfusion experts optimized for different conditions. The subfusion experts utilize wavelet transforms to perform frequency decoupling on the input multimodal features, and then carry out personalized fusion on the decoupled

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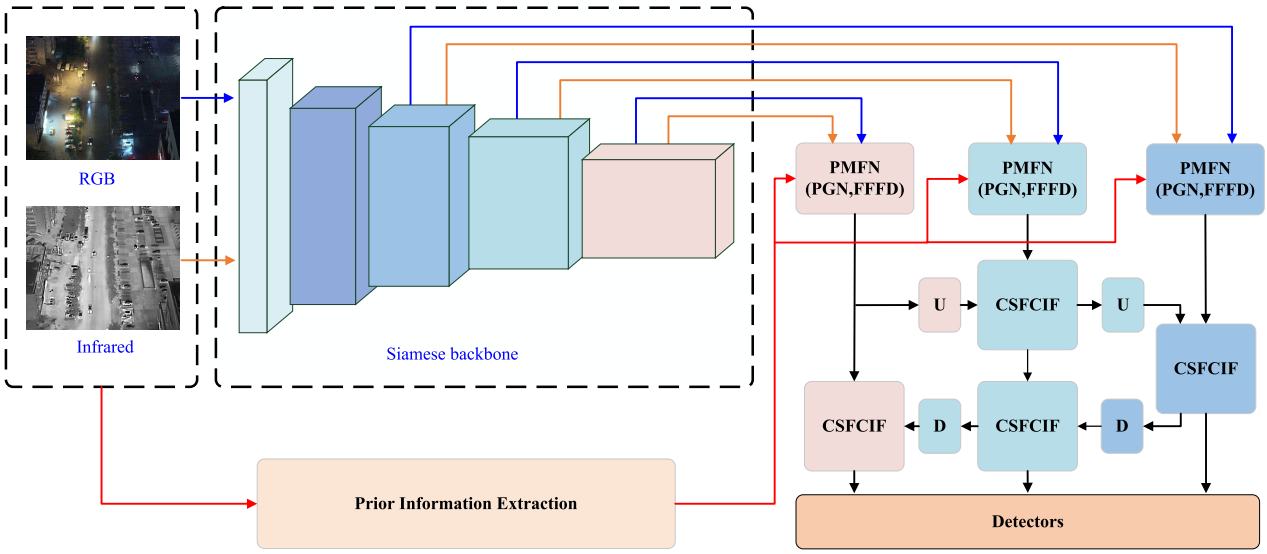


Fig. 2. Framework of multimodal remote sensing object detection based on prior-enhanced MoE fusion network. "PMFN" represents the prior information-enhanced MoE fusion network. "FFFD" represents the subfusion expert network in the MoE fusion network, specifically the feature fusion network based on frequency decoupling. "CSFCIF" refers to the cross-scale feature channel interleaving fusion module. "U" and "D" indicate the upsampling and downsampling modules, respectively.

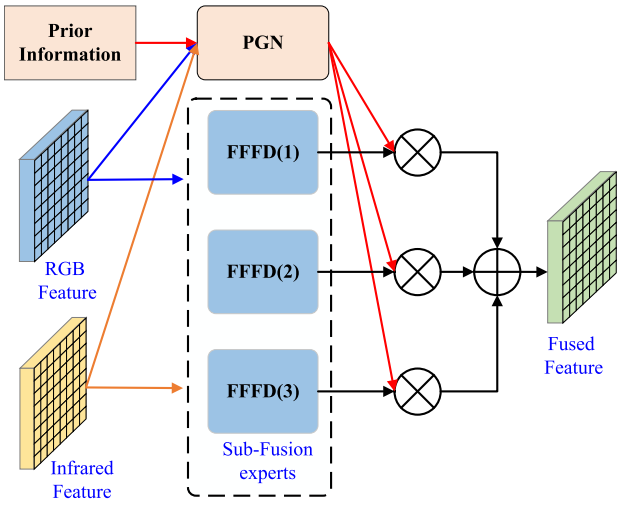


Fig. 3. PMFN. "FFFD" represents the subfusion expert network within the PMFN, which serves as a feature fusion network based on frequency decoupling. "PGN" represents the prior information-assisted dynamic gating network.

features' different frequency subbands. Finally, the outputs from the subfusion experts are weighted and fused to generate environment-adaptive multimodal features

$$F_{\text{fuse}} = \sum_{i=1}^{n} g_i \cdot e_i(F_{\text{vis}}, F_{\text{ir}}) \tag{1}$$

where F_{fuse} denotes the fused features of infrared and visible images, g_i represents the gating weight for expert i, e_i corresponds to the subfusion expert i, and F_{rgb} and F_{ir} represent the visible image features and infrared image features, respectively.

Prior information are a crucial channel for environmental perception, while environmental factors are key determinants affecting the robustness of fusion detection methods. Moreover, the features of multimodal images play a significant role in both environmental perception and the robustness of detection algorithms. Therefore, the MoE fusion network designed in this article does not merely partition the environment explicitly based on prior information alone. Instead, it integrates prior information with the feature information extracted from infrared and visible images, adhering to the principle of jointly driven knowledge and data to achieve an implicit partitioning of the environment.

B. Prior Information-Assisted Dynamic Gating Network

The role of the PGN is to sense environmental factors, such as illumination and weather, and to assign weights to the subfusion experts optimized for different environments, thereby laying the foundation for enhancing the robustness of the detection algorithm. The gating network receives information from two sources: one is prior information (grayscale histogram statistics and information entropy metrics), and the other is the current features of the infrared and visible images (Fig. 4).

The prior information include the grayscale distribution of the visible image Hist$\{h_1, h_2, \ldots, h_n\}$, the information entropy of the infrared image EN_{ir}, and the information entropy of the visible image EN_{vis}. These prior information serve as a numerical description of the image's global information and also act as an important basis for assessing the overall environment. For example, analyzing the grayscale distribution of a visible image can effectively infer the approximate lighting conditions of the environment. Notably, even under high visible image brightness, there remains the possibility of low visibility, which can reduce the information entropy metrics of the visible image. Since low visibility, which is often caused by factors such as haze, has a minimal impact on infrared images, the combined consideration of brightness and the information entropy metrics from both modalities can

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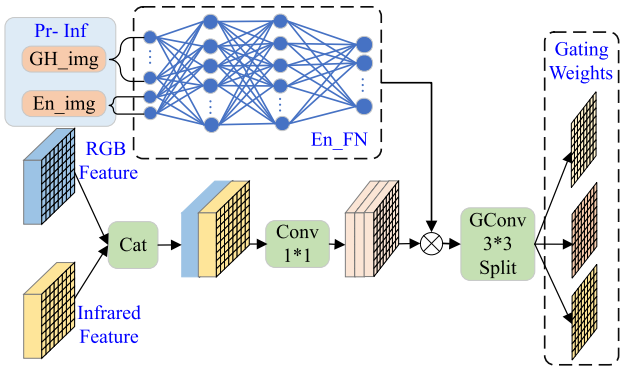


Fig. 4. Block diagram of PGN. Pr-Inf represents prior information. GH_img denotes the grayscale histogram statistics of the visible image. En_img represents the information entropy of both infrared and visible images. Gconv denotes grouped convolution, where the number of groups matches the number of experts, ensuring that the final number of input channels equals the number of experts. "Split" represents channelwise splitting, with the result serving as the gating weights for each subexpert. "En_FN" denotes the prior information encoder, which is composed of a fully connected network.

be used to determine whether a visible image is captured under low-visibility conditions.

In Hist$\{h_1, h_2, \ldots, h_n\}$, n represents the number of grayscale intervals. Since only a rough estimate of the grayscale distribution is required in this study, n is set to 8. The entropy calculations for the infrared and visible images, $\mathrm{EN_{ir}}$ and $\mathrm{EN_{vis}}$, are given by

$$\mathrm{EN_{ir}} = -\sum_{i=0}^{m} p_i^{\mathrm{ir}} \mathrm{Log}\left(p_i^{\mathrm{ir}}\right) \tag{2}$$

$$\mathrm{EN_{vis}} = -\sum_{i=0}^{m} p_i^{\mathrm{vis}} \mathrm{Log}\left(p_i^{\mathrm{vis}}\right) \tag{3}$$

where m represents the number of grayscale intervals, which is set to 127 in this study.

Within the gated network, the current features from the input infrared and visible images are first concatenated along the channel dimension, enabling both modalities to collaboratively contribute to environmental perception. Subsequently, convolution is applied to fuse these concatenated features, resulting in the mixed feature $f^g_{\mathrm{ir_vis}}$. This process enhances the interchannel correlation and overall integrity of the features, thereby facilitating the subsequent incorporation of prior information to effectively support the MoE fusion network. Meanwhile, multimodal image features provide local detailed information for environmental perception

$$f^g_{\mathrm{ir_vis}} = \mathrm{Conv}^{1\times1}(\mathrm{Cat}(f_{\mathrm{ir}}, f_{\mathrm{vis}})) \tag{4}$$

where $\mathrm{Conv}^{1\times1}$ represents the convolutional layer with a kernel size of 1, Cat denotes the channel concatenation, f_{ir} refers to the infrared image features, and f_{vis} represents the visible image features.

Meanwhile, the grayscale distribution and other prior information from the visible image are mapped to a latent space through a fully connected network, generating the comprehensive prior information Pr. By integrating prior knowledge and transforming it into a highly abstract and encoded form Pr, the model can leverage these additional, guiding prior cues to assist the gating network in making informed decisions

$$\mathrm{Pr} = \delta(\mathrm{FN}(\mathrm{Cat}(\mathrm{Hist}, \mathrm{EN_{ir}}, \mathrm{EN_{vis}}))) \tag{5}$$

where δ represents the sigmoid function, while FN denotes the fully connected network with two hidden layers.

Subsequently, the integrated prior information Pr is combined with the mixed feature $f^g_{\mathrm{ir_vis}}$ to obtain $f^p_{\mathrm{ir_vis}}$. Building upon $f^g_{\mathrm{ir_vis}}$, $f^p_{\mathrm{ir_vis}}$ incorporates prior information to ensure that the current features and prior knowledge work synergistically, thereby providing a more instructive informative input for generating subsequent gating weights. Unlike explicit environmental classification approaches, our design adopts an implicit modeling strategy, where the gating network learns to exploit the statistical characteristics of the input data to adjust its behavior adaptively. This implicit mechanism enables the model to dynamically respond to varying environmental conditions (e.g., lighting or visibility changes) without requiring manually defined environmental labels or rules, thus enhancing robustness and generalization in complex real-world scenes

$$f^p_{\mathrm{ir_vis}} = f^g_{\mathrm{ir_vis}} \cdot \mathrm{Pr}. \tag{6}$$

To determine the specialized fusion responsibilities of individual subexperts, the final spatial-type gating weights are derived through dimensional reduction of $f^p_{\mathrm{ir_vis}}$ using grouped convolutional operations. Specifically, the feature dimensionality is compressed through groupwise convolutional processing, where the number of groups corresponds to the cardinality of subfusion expert networks, thereby generating spatial-type gating weights G

$$G = G\mathrm{Conv}_n\left(f^p_{\mathrm{ir_vis}}\right) \tag{7}$$

where n represents the number of output channels, which is also the number of experts. Through backpropagation, the parameters of the grouped convolution are dynamically adjusted based on the task loss, ensuring that the weight allocation for each subfusion expert is precisely aligned with environmental features (such as illumination, noise, and scene complexity) and task requirements.

Different regions within the same image may experience varying environmental conditions (e.g., strong light on the left side and shadow on the right side). Traditional gating mechanisms struggle to capture such spatial variations. In contrast, a spatial gating mechanism can dynamically adjust the contributions of different subfusion models based on the local environment of each pixel. Therefore, by introducing a spatial dimension dynamic routing mechanism, the MoE fusion network gains pixel-level scene adaptation capability.

C. Frequency-Decoupled Feature Fusion Network

In multimodal remote sensing images, frequency components reflect distinct types of information. The low-frequency subbands typically retain the global structure and intensity distribution, such as roads, vehicles, and building outlines. In infrared images, this reflects large-scale thermal patterns, while in visible images, it reflects general scene layout and lighting. On the other hand, the high-frequency subbands capture fine-grained details and texture edges.

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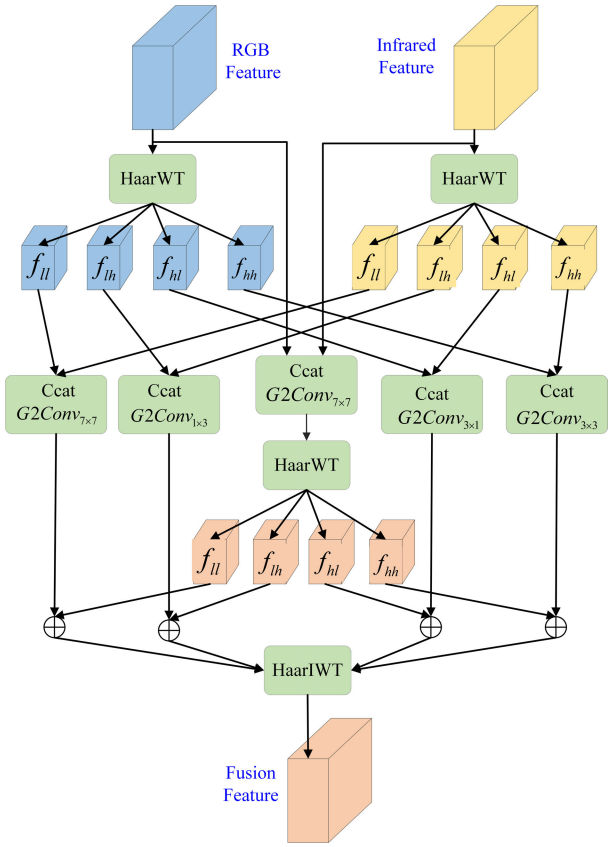


Fig. 5. Framework of infrared and visible image feature fusion based on frequency decoupling. "HaarWT" denotes the Haar wavelet transform. "Ccat" represents channelwise interleaved concatenation of features. "G2Conv" refers to grouped convolution, where each group processes two input channels.

Infrared high-frequency components highlight sharp temperature transitions, such as heat boundaries of vehicles, while visible high-frequency components provide rich texture cues like pedestrian contours. These complementary characteristics motivate our frequency-specific fusion strategy, enabling the network to effectively preserve and utilize both semantic structures and discriminative details for object detection.

Therefore, subfusion experts in the MoE fusion network are implemented using the feature fusion network based on frequency decoupling (FFFD) network. The core architecture of this network comprises two key components: a frequency subband fusion branch based on wavelet transform and a spatial fusion branch. The frequency subband fusion branch decomposes infrared and visible image features into different frequency subbands using the wavelet transform and then fuses each frequency subband separately. Decoupling these frequency components allows for the design of more appropriate fusion strategies tailored to the characteristics of each band, while also preventing mutual interference during the fusion process. Meanwhile, the spatial fusion branch directly fuses the original, undecomposed image features, thereby reinforcing the synergistic interaction among the different frequency subband information (Fig. 5).

Specifically, the frequency subband fusion branch based on wavelet transform applies the Haar wavelet transform to the features of the input infrared and visible images separately

$$\left(f_{\text{ir}}^{\text{ll}}, f_{\text{ir}}^{\text{lh}}, f_{\text{ir}}^{\text{hl}}, f_{\text{ir}}^{\text{hh}}\right) = \text{HaarWT}(f_{\text{ir}}) \tag{8}$$

$$\left(f_{\text{vis}}^{\text{ll}}, f_{\text{vis}}^{\text{lh}}, f_{\text{vis}}^{\text{hl}}, f_{\text{vis}}^{\text{hh}}\right) = \text{HaarWT}(f_{\text{vis}}) \tag{9}$$

where HaarWT() denotes the Haar wavelet transform applied to the input features, f^{ll} represents the low-frequency subband of the features, f^{lh} represents the horizontal high-frequency subband, f^{hl} represents the vertical high-frequency subband, and f^{hh} represents the diagonal high-frequency subband.

The frequency subbands of the infrared and visible image features, obtained through wavelet decomposition, are fused separately for each component

$$f_{\text{ir_vis}}^{\text{ll}} = \text{G2Conv}_{7\times7}\left(C\text{cat}\left(f_{\text{ir}}^{\text{ll}}, f_{\text{vis}}^{\text{ll}}\right)\right) \tag{10}$$

$$f_{\text{ir_vis}}^{\text{lh}} = \text{G2Conv}_{1\times3}\left(C\text{cat}\left(f_{\text{ir}}^{\text{lh}}, f_{\text{vis}}^{\text{lh}}\right)\right) \tag{11}$$

$$f_{\text{ir_vis}}^{\text{hl}} = \text{G2Conv}_{3\times1}\left(C\text{cat}\left(f_{\text{ir}}^{\text{hl}}, f_{\text{vis}}^{\text{hl}}\right)\right) \tag{12}$$

$$f_{\text{ir_vis}}^{\text{hh}} = \text{G2Conv}_{3\times3}\left(C\text{cat}\left(f_{\text{ir}}^{\text{hh}}, f_{\text{vis}}^{\text{hh}}\right)\right) \tag{13}$$

where Ccat represents the channelwise interleaved concatenation of infrared and visible image features and $\text{G2Conv}_{7\times7}$ denotes the grouped convolution with a kernel size of 7×7, where each group consists of two channels, with one channel from the infrared image features and the other from the visible image features. Similarly, $\text{G2Conv}_{1\times3}$, $\text{G2Conv}_{3\times1}$ and $\text{G2Conv}_{3\times3}$ represent grouped convolutions with kernel sizes of 1×3, 3×1, and 3×3, respectively.

Since the backbone network of this method adopts a Siamese network structure, the corresponding channels in the infrared and visible image features exhibit high semantic correlation. Therefore, leveraging grouped convolution to fuse the corresponding channels of infrared and visible image features can fully exploit the advantages brought by the Siamese backbone structure.

Studies on large-kernel convolutions, such as ConvNeXt [40], RepLKNet [41], and PeLK [42], have demonstrated that a large receptive field enhances remote sensing object detection performance. The low-frequency subband primarily contains the overall structure and large-scale features of an image. Using large convolutional kernels for fusion allows for the integration of broader contextual information while preserving globally important features. Research such as HFC-EGCNN [43] has shown that high-frequency information play a crucial role in object detection tasks. The horizontal high-frequency subband mainly captures detail variations in the horizontal direction. Employing a 1×3 convolutional kernel for fusion helps retaining the texture details of infrared and visible image features along the horizontal axis while reducing interference from other directions. Similarly, the vertical high-frequency subband reflects detail variations in the vertical direction, and using a 3×1 convolutional kernel effectively integrates these vertical details. The diagonal high-frequency subband contains diagonal direction details, which are relatively complex. A 3×3 convolutional kernel balances local detail extraction and contextual information by covering a moderate region, making it suitable for fusing infrared and visible image features in diagonal directions.

For the spatial-domain fusion branch, the input infrared and visible image features are first channelwise interleaved

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and concatenated, followed by direct fusion to obtain the spatial-domain fused features $f_{\text{vis_ir}}^{\text{sp}}$:

$$f_{\text{vis_ir}}^{\text{sp}} = \text{G2Conv}_{7\times7}\left(C\text{cat}\left(f_{\text{ir}}, f_{\text{vis}}\right)\right). \tag{14}$$

Subsequently, the fusion results from both branches are further integrated to obtain the final fused features $f_{\text{ir_vis}}^{f}$

$$\left(f_{\text{ir_vis}}^{\text{sp_ll}}, f_{\text{ir_vis}}^{\text{sp_lh}}, f_{\text{ir_vis}}^{\text{sp_hl}}, f_{\text{ir_vis}}^{\text{sp_hh}}\right) = \text{HaarWT}\left(f_{\text{ir_vis}}^{\text{sp}}\right) \tag{15}$$

$$f_{\text{ir_vis}}^{f} = \text{HaarIWT}\left(\left(f_{\text{ir_vis}}^{\text{ll}} + f_{\text{ir_vis}}^{\text{sp_ll}}\right)\left(f_{\text{ir_vis}}^{\text{lh}} + f_{\text{ir_vis}}^{\text{sp_lh}}\right)\left(f_{\text{ir_vis}}^{\text{hl}} + f_{\text{ir_vis}}^{\text{sp_hl}}\right)\left(f_{\text{ir_vis}}^{\text{hh}} + f_{\text{ir_vis}}^{\text{sp_hh}}\right)\right) \tag{16}$$

where HaarWT() denotes the inverse Haar wavelet transform.

The information contained in different frequency subbands of different modality images varies in importance. Selecting an appropriate fusion strategy can help avoid introducing noise during the fusion process. For low-frequency subband fusion, large convolutional kernels can smooth out small fluctuations caused by differences between the two image sensors. In contrast, for high-frequency subband fusion, small convolutional kernels can focus on refining true details, thereby improving the quality of the fused features. Meanwhile, the spatial-domain branch can be regarded as a hybrid-frequency fusion mechanism, enhancing the connection between high-frequency and low-frequency informations.

D. Cross-Scale Feature Channel Interleaved Fusion (CSFCIF) Module

Within object detection architectures, the path aggregation feature pyramid network (PaFPN) [44], [45] plays an indispensable role. In past practices, PaFPN has primarily employed two methods for fusing features of different scales. The first approach involves upsampling deep features (a bottom-up pathway) so that their resolution matches that of shallow features, followed by channelwise concatenation for fusion. The second approach downsamples shallow features (a top-down pathway) to align their scale with deep features, after which they are fused via channel concatenation. However, these traditional concatenation-based fusion methods tend to weaken the spatial correlations among features, and the subsequent dimensionality reduction process introduces considerable computational overhead, thereby increasing both the required computational resources and the runtime. In contrast, the CSFCIF strategy proposed in this article aligns shallow and deep features via upsampling or downsampling, then interleaves them along the channel dimension—breaking away from the conventional direct concatenation approach—to enhance the exchange of information between features of different scales. Subsequently, grouped convolution is applied to the interleaved features for deep fusion, which improves the efficiency of the fusion process while maintaining high fusion quality (Fig. 6). The detailed scheme is as follows.

The bottom-up pathway

$$f = \text{G3Conv}\left(\text{C2cat}\left(f_j, Up\left(f_{j-1}\right)\right)\right) \tag{17}$$

where f_j denotes the input feature at scale j while f_{j-1} represents the feature at the next scale relative to f_j. The height and width of f_{j-1} are half those of f_j, but the number of channels in f_j is half that of f_{j-1}. $Up()$ is the upsampling function. G3Conv() represents the grouped convolution in which each group comprises three channels, two of which come from the upsampled features f_{j-1}. C2cat() denotes the cross-channel concatenation, meaning that two channels from the upsampled f_{j-1} features are concatenated with one channel from f_j.

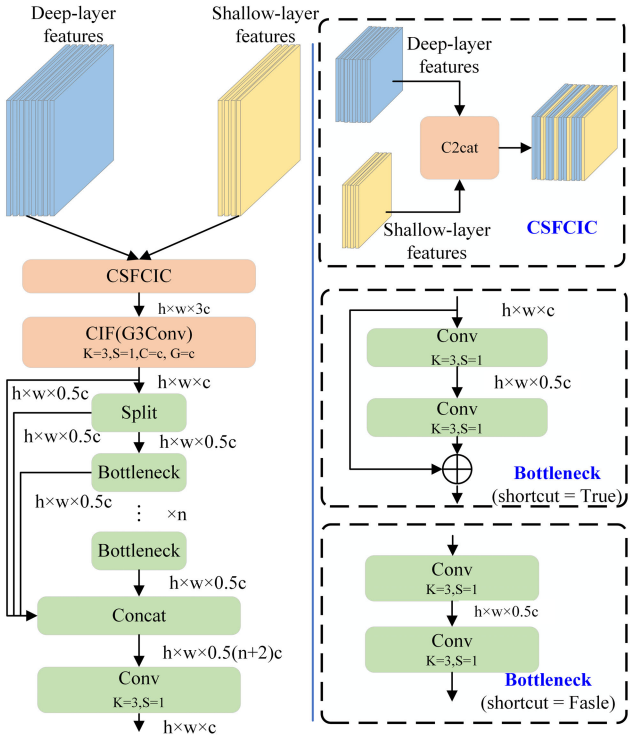


Fig. 6. Cross-scale feature channel interleaved concatenation and fusion structure. "CSFCIC" represents the cross-scale feature channel interleaved concatenation module, while "CIF" denotes the channel interleaved fusion module.

The top-down pathway

$$f = \text{G3Conv}(\text{C2cat}(f_{i-1}, Dp(f_i))) \tag{18}$$

where $Dp()$ represents the downsampling function.

E. Loss Function

The loss function in this article follows the design of conventional object detection algorithms, retaining their inherent localization loss and classification loss [45]. The key difference lies in the addition of a self-supervised expert differentiation loss derived from the MoE model. Thus, the loss function is defined as

$$L_{\text{all}} = L_{\text{cls}} + L_{\text{dfl}} + L_{\text{box}} + L_{\text{dev}} \tag{19}$$

where L_{cls} represents the inherent classification loss in the object detection algorithm, while L_{dfl} and L_{box} denote the inherent localization losses in the object detection framework.

Although gated networks possess the capability to guide different experts toward optimizing in different directions, the potential for overlapping optimization directions may still arise due to the same network architectures employed by each

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expert. To mitigate the redundancy in the fusion capabilities of distinct expert models, a self-supervised expert differentiation diversity loss (L_{dev}) is, thus, designed

$$L_{\text{div}} = \frac{1}{2c} \sum_{s\in(s_1,s_2,s_3)} \left(\sigma\left(\text{Sim}_{\text{cp}}\left(E^s\right)\right) + \sigma\left(\text{Sim}_{\text{cp}}\left(E^s\right)\right)\right) \quad (20)$$

where s represents the features at different scales, c denotes the number of scales, and σ refers to the LeakyReLU activation function. The primary purpose of using LeakyReLU is to constrain the positively correlated components in the diversity loss while allowing for the presence of negative correlations among different experts. However, to prevent extreme cases of complete negative correlation, LeakyReLU also imposes a slight constraint on negatively correlated components. $\text{Sim}_{\text{cp}}()$ denotes the channelwise sparse similarity calculation function, while $\text{Sim}_{\text{sp}}()$ represents the spatial sparse similarity calculation function. $E^s = \left\{E_1^s, E_2^s, \ldots, E_m^s\right\}$ refers to the expert outputs, where m represents the number of experts, which is set to 3 in this work

$$\text{Sim}_{\text{sp}}\left(E^s\right) = \frac{2}{m(m-1)} \sum_{1\le i<j\le m} \cos\left(\text{avg_sp}\left(E_i^s\right)\text{avg_sp}\left(E_j^s\right)\right) \quad (21)$$

$$\text{Sim}_{\text{cp}}\left(E^s\right) = \frac{2}{m(m-1)} \sum_{1\le i<j\le m} \cos\left(\text{avg_cp}\left(E_i^s\right)\text{avg_cp}\left(E_j^s\right)\right) \quad (22)$$

where cos() represents the cosine similarity calculation function, and avg_sp() denotes the global average pooling in the spatial dimension, while avg_cp() refers to global average pooling in the channel dimension. Sparse similarity calculation prevents absolute decorrelation between different experts.

IV. Experiments

A. Datasets

In this article, our proposed method is trained and validated on the DroneVehicle [5] and RGBT-Tiny [21] datasets.

The DroneVehicle dataset is a comprehensive UAV-based remote sensing dataset featuring both visible and infrared imagery, specifically designed for vehicle detection tasks. It includes 28 439 paired images, annotated with oriented bounding boxes across five categories: car, bus, truck, van, and freight car, totaling 953 087 labeled instances. Captured under day and night conditions, the dataset covers urban roads, residential areas, highways, and other scenarios, with images taken from various heights and angles. For robust evaluation, the dataset is partitioned into training (17 990 pairs), validation (1469 pairs), and testing (8980 pairs) subsets, ensuring balanced category distributions and consistent sampling across different perspectives.

The RGBT-Tiny dataset is a large-scale benchmark designed for small object detection in visible–infrared images, primarily captured from UAV-based remote sensing. It consists of 115 paired sequences, 93k frames, and 1.2M manually annotated objects across seven categories (ship, car, cyclist, pedestrian, bus, drone, and plane) and eight diverse scene types (country road, city road, bridge, lake, sea, sky, airport, playground). Over 81% of the objects are smaller than 16×16 pixels. All paired visible–infrared images are provided at a uniform resolution of 512×640 pixels. In the RGBT-tiny dataset, objects are annotated with vertical bounding boxes.

B. Implementation Detail

All experiments in this article were implemented using PyTorch on an NVIDIA GeForce RTX A6000 GPU with Ubuntu 22 and CUDA 11.7. For the DroneVehicle dataset, we employed a baseline method (YOLOv8s-obb [45]) that utilizes a Siamese backbone and additive fusion approach. The method was trained for 100 epochs using the SGD optimizer with an initial learning rate of 0.01 and a weight decay of 0.01, with input images sized at 640×640 pixels. In contrast, due to the small object sizes in the RGBT-tiny dataset, we adopted a Siamese backbone and additive fusion-based YOLOv8s-p2 [45] as the baseline. This method was trained for 30 epochs with the same SGD optimizer settings (initial learning rate of 0.01 and weight decay of 0.01), and output images were also sized at 640×640 pixels. Evaluation protocols for the DroneVehicle dataset conformed to the methodology established in [5], where performance was quantified using mean average precision (mAP) at an IoU threshold of 0.5. For the RGBT-tiny dataset, detection accuracy was measured following the COCO-compliant evaluation pipeline detailed in [21], with an average precision (AP) computed across a range of IoU thresholds from 0.5 to 0.95 at 0.05 intervals as specified in [50].

C. Comparisons With State-of-the-Art Methods in DroneVehicle

To thoroughly evaluate the performance of the proposed MDPMFN method, we conducted extensive comparisons with multiple state-of-the-art object detection algorithms (Fig. 7). Specifically, the evaluation covers ten advanced visible–infrared detection methods, including representative models such as M2FP [49] and C2Former, along with five single model detection algorithms as benchmark references. Detailed comparative results are summarized in Table I.

The experimental results indicate that the proposed MDPMFN method attains an mAP score of 81.26, surpassing all compared methods. It achieves a 1.86% improvement in mAP over the second-best method (DMM), demonstrating robust performance in complex scenarios and a 2.56% gain compared to M2FP, which reflects the advantages of its frequency-domain fusion and detection framework for visible–infrared data. Furthermore, MDPMFN shows 4.33% and 7.03% mAP improvements over FFODNet and C2Former, respectively, indicating enhanced fusion feature representation. For critical object categories such as "car," "bus," and "truck," the method consistently yields higher mAP scores than existing approaches. These results demonstrate that the PMFN effectively enhances the ability of remote sensing object detection algorithms to handle complex environments.

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TABLE I

COMPARISON OF MDPMFN WITH STATE-OF-THE-ART METHODS ON THE DRONEVEHICLE DATASET. THE TERMS "RGB" AND "IR" DENOTE DETECTION OUTCOMES FROM VISIBLE IMAGES AND INFRARED IMAGES, RESPECTIVELY, WHILE "FUSION" REFLECTS COMBINED DETECTION OUTCOMES FROM FUSED INFRARED AND VISIBLE IMAGES. RED-COLORED NUMBERS HIGHLIGHT OPTIMAL PERFORMANCE METRICS, WITH BOLDFACE ENTRIES CORRESPONDING TO SECONDARY-BEST RESULTS ACROSS COMPARATIVE ANALYSES

Detectors	Modality	car	bus	truck	freight car	van	mAP
Faster R-CNN [10]	RGB	67.88	66.98	38.59	26.31	23.2	44.59
RoITransformer [11]	RGB	68.13	70.55	44.17	29.08	27.64	47.91
ReDet [13]	RGB	69.48	77.37	47.87	31.46	29.03	51.04
Gliding Vertex [12]	RGB	75.77	68.05	46.08	33.75	38.72	52.48
R3Det [16]	RGB	79.30	76.00	42.20	24.50	28.50	50.10
Faster R-CNN [10]	IR	88.63	77.92	42.51	35.16	28.52	54.55
RoITransformer [11]	IR	88.85	79.48	51.53	41.49	34.39	59.15
ReDet [13]	IR	89.47	79.89	53.95	42.82	36.56	60.54
Gliding Vertex [12]	IR	89.15	78.75	59.72	42.95	43.88	62.89
R3DET [16]	IR	89.50	87.10	48.30	16.60	39.90	62.30
UA-CMDet [5]	Fusion	87.51	87.08	60.70	46.80	37.95	64.01
TSFADet [46]	Fusion	89.23	88.08	72.00	54.15	48.77	70.44
CLGNet [3]	Fusion	-	-	-	-	-	75.10
CALNet [23]	Fusion	90.30	89.11	76.15	62.97	58.46	75.39
FFODNet [28]	Fusion	-	-	-	-	-	76.93
DMM [47]	Fusion	90.40	89.90	**79.80**	68.20	68.60	**79.40**
CAGTDet [48]	Fusion	90.82	90.46	69.65	66.28	55.62	74.54
FMCFNet [37]	Fusion	90.33	89.35	77.71	65.65	60.71	76.75
C2Former [4]	Fusion	90.21	89.81	68.26	64.42	58.47	74.23
M2FP [49]	Fusion	**95.70**	**92.10**	76.20	64.70	**64.70**	78.70
MDPMFN(Our)	Fusion	97.95	95.50	80.87	**67.68**	64.33	81.26

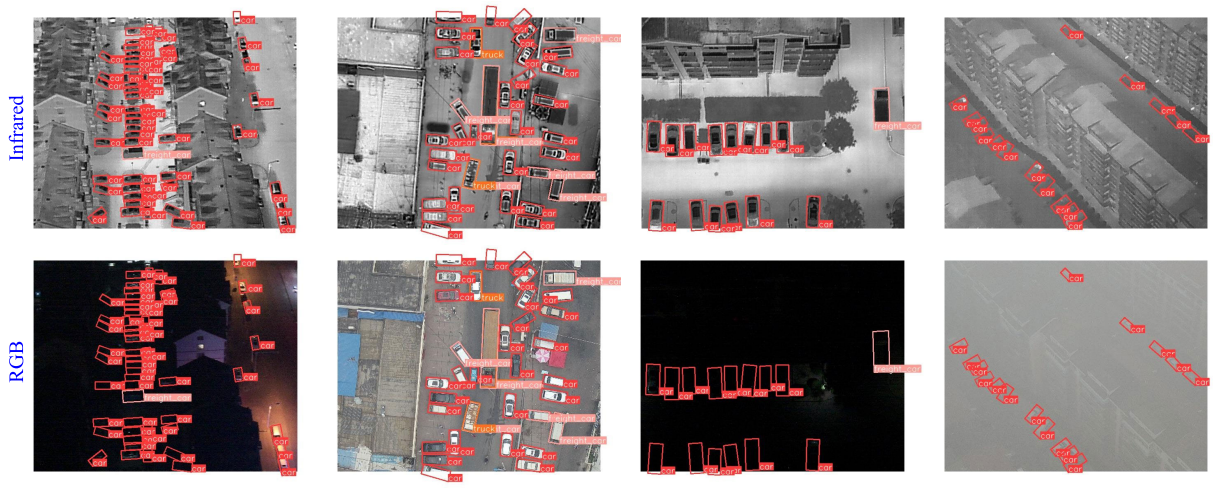


Fig. 7. Results of MDMPFN fusion detection on the DroneVehicle dataset, shown on the infrared and visible images, respectively.

D. Comparisons With State-of-the-Art Methods in RGBT-Tiny

To thoroughly assess the performance of the proposed MDPMFN method, we conducted extensive comparisons with multiple state-of-the-art object detection algorithms on the RGBT-tiny dataset (Fig. 8). The evaluated methods include single modal detectors (e.g., Faster R-CNN and SABL [51] for infrared or visible images object detection) and multimodal detectors (e.g., UA-CMDet and ProbEn-early for visible–infrared images object detection). Comparative results are summarized in Table II.

The experimental results demonstrate that the proposed MDPMFN method achieves an AP score of 37.0 on the RGBT-tiny dataset, outperforming all evaluated single modal (e.g., Faster RCNN and SABL) and multimodal (e.g., UA-CMDet and ProbEn-early) detectors. Specifically, it surpasses the top-performing infrared-only method RFLA by 3.2% and the visible-only RFLA by 4.9% in AP, validating its effectiveness in leveraging complementary visible–infrared fusion strategies. Compared to existing multimodal approaches, MDPMFN exhibits notable improvements, underscoring the advantages of its expert mixture mechanism. Furthermore, the method excels in multiscale detection, achieving 3.2% and 5.0% gains in AP_t^s (tiny objects) and AP_s^s (small objects) over suboptimal method, respectively, while maintaining competitive performance for extremely tiny objects, medium and large objects (AP_{et}^s, AP_m^s, and AP_l^s). These results highlight its robustness in handling diverse object scales and complex multimodal scenarios.

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TABLE II

COMPARISON OF MDPMFN WITH STATE-OF-THE-ART METHODS ON THE RGBT-TINY DATASET. THE TERMS "RGB" AND "IR" DENOTE DETECTION OUTCOMES FROM VISIBLE IMAGES AND INFRARED IMAGES, RESPECTIVELY, WHILE "FUSION" REFLECTS COMBINED DETECTION OUTCOMES FROM FUSED INFRARED AND VISIBLE IMAGES. RED-COLORED NUMBERS HIGHLIGHT OPTIMAL PERFORMANCE METRICS, WITH BOLDFACE ENTRIES CORRESPONDING TO SECONDARY-BEST RESULTS ACROSS COMPARATIVE ANALYSES. AP^s_{ET}, AP^s_t,AP^s_s, AP^s_m, AND AP^s_l REPRESENT THE AP VALUES FOR EXTREMELY TINY, TINY, SMALL, MEDIUM, AND LARGE OBJECTS, RESPECTIVELY

Detectors	Modality	AP^s_{et}	AP^s_t	AP^s_s	AP^s_m	AP^s_l	AP
Faster R-CNN [10]	IR	21.9	35.2	45.0	41.4	**80.0**	29.5
SABL [51]	IR	21.8	34.8	47.2	**42.6**	90.0	29.3
FCOS [52]	IR	14.0	17.4	38.6	29.0	90.0	16.9
VarifocalNet [53]	IR	20.2	34.7	**51.4**	46.8	90.0	30.4
Deformable DETR [54]	IR	21.4	31.6	48.1	37.6	70.0	28.0
Sparse R-CNN [55]	IR	18.0	22.4	31.6	31.0	40.0	20.6
QueryDet [56]	IR	19.3	31.0	40.1	21.3	40.0	25.1
RFLA [57]	IR	27.5	**37.6**	49.8	38.8	60.0	**33.8**
ACM [19]	IR	21.7	25.0	22.1	11.4	40.0	18.2
ALCNet [20]	IR	13.9	15.3	17.8	7.2	19.3	12.2
DNAnet [18]	IR	14.9	13.8	31.3	14.3	0.0	13.6
Faster R-CNN [10]	RGB	24.3	30.1	44.2	22.0	65.0	28.8
SABL [51]	RGB	24.0	31.0	46.6	25.5	77.6	29.6
FCOS [52]	RGB	15.3	18.8	32.4	20.0	45.2	17.5
VarifocalNet [53]	RGB	22.1	27.8	45.0	27.9	**80.0**	26.9
Deformable DETR [54]	RGB	25.2	27.2	43.5	22.9	65.0	28.2
Sparse R-CNN [55]	RGB	18.8	19.5	33.4	14.9	40.1	19.2
QueryDet [56]	RGB	19.2	25.1	37.0	20.3	30.0	23.6
RFLA [57]	RGB	**26.8**	30.4	45.3	29.4	50.0	32.1
ACM [19]	RGB	8.8	5.7	5.0	0.0	0.0	1.2
ALCNet [20]	RGB	19.6	9.3	2.0	0.0	0.0	2.0
DNAnet [18]	RGB	1.0	4.4	6.2	14.5	0.0	2.6
UA-CMDet [5]	Fusion	10.8	24.1	35.0	38.5	9.7	20.0
ProbEn-early [21]	Fusion	12.3	20.2	32.3	22.2	76.7	17.0
ProbEn-middle [21]	Fusion	21.8	35.3	44.1	37.5	70.0	29.3
MDPMFN(Our)	Fusion	25.6	40.8	56.4	41.1	**80.0**	37.8

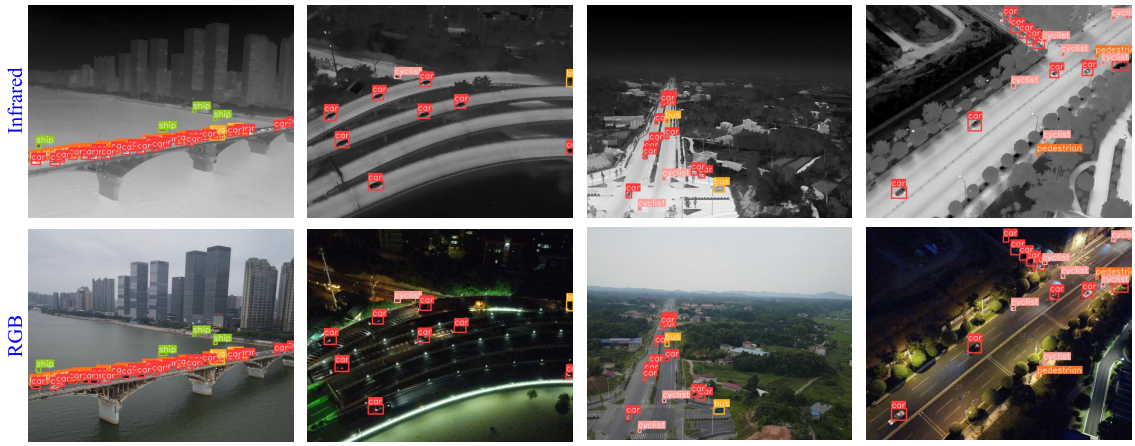


Fig. 8. Results of MDPMFN fusion detection on the RGBT-tiny dataset, shown on the infrared and visible images, respectively.

E. Ablation Study

This section presents ablation experiments to quantify the individual contributions of each component in our model. In addition, it examines the impact of prior information, as well as the effects of low- and high-frequency informations through ablation analysis. The evaluations are carried out using the test subset of the DroneVehicle dataset.

1) Ablation Study on Key Module: Table III presents an ablation analysis of the key modules in MDPMFN. The red numbers indicate the best performance, while the bold numbers represent the second-best performance.

According to the data in Table III, after introducing the FFFD, the mAP metric improved by 3.47%. This result strongly demonstrates that the differentiated fusion strategy designed for the characteristics of different frequency subbands effectively preserves critical frequency information from the source features, thereby enhancing object detection performance. Moreover, since the fusion strategy for different frequency subbands is primarily implemented using depthwise

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TABLE III

RESULTS OF THE ABLATION ANALYSIS OF THE KEY MODULES IN MDPMFN. ✓ DENOTES THE INCLUSION OF THE RESPECTIVE COMPONENT

Methods	FFFD	PMFN	CSFCIF	mAP	FLOPs
Baseline				74.94	**42.2**
Baseline+FFFD	✓			78.41	42.3
Baseline+FFFD+PMFN	✓	✓		81.49	43.3
MDPMFN	✓	✓	✓	**81.26**	41.5

TABLE IV

IMPACT OF REMOVING PRIOR INFORMATION ON DETECTION ACCURACY (MDPMFN_NO_PR: MODEL WITHOUT PRIOR INFORMATION INTEGRATION; MDPMFN: FULL MODEL WITH PRIOR INFORMATION)

	MDPMFN_no_pr	MDPMFN
car	**97.86**	97.95
bus	95.69	**95.50**
truck	81.13	**80.87**
freight car	**66.06**	67.68
van	**62.08**	64.33
mAP	**80.57**	81.26

separable convolutions, spatially separated convolutions, and grouped convolutions, the fusion expert network incurs a relatively low computational cost. Based on the data in Table III, incorporating the PMFN resulted in a 2.61% increase in mAP. This result clearly demonstrates that the PMFN achieves environment-adaptive fusion of multimodal image features, thereby improving object detection performance. Furthermore, the data presented in Table III shows that integrating the CSFCIF module into the detection algorithm significantly reduces computational load by up to 1.8 FLOPs, while the mAP accuracy decreases by only 0.23%. This experimental result strongly demonstrates that the CSFCIF module can effectively lower computational complexity while maintaining detection accuracy at a relatively stable level.

2) Ablation Study on Prior Information Quantification Mechanism: We conducted ablation experiments by removing the prior information quantification mechanism from the proposed MDPMFN model, resulting in a variant denoted as MDPMFN_no_pr. The performance comparison between MDPMFN and MDPMFN_no_pr is summarized in Table IV, including detailed evaluations across object categories and mAP. The exclusion of prior information quantification leads to a noticeable decline in overall detection performance, with the mAP dropping from 81.26 (MDPMFN) to 80.57 (MDPMFN_no_pr). Although categories, such as car and bus, exhibit only minor fluctuations, the impact is more substantial for freight car and van, with reductions of 1.62% and 2.25%, respectively. These results underscore the critical role of our prior information quantification mechanism. While its influence varies across object categories, the consistent drop in mAP validates its contribution to enhancing the model's adaptability in dynamic environments through the integration of histogram statistics and entropy metrics within a gated network.

TABLE V

ABLATION STUDY ON THE CONTRIBUTION OF FREQUENCY SUBBANDS TO MDPMFN PERFORMANCE

	low-frequency	high-frequency	mAP
MDPMFN(no_HF)	✓		78.73
MDPMFN(no_LF)		✓	**80.63**
MDPMFN	✓	✓	81.26

3) Ablation Study on Cross-Modal Frequency-Specific Fusion: To assess the impact of low-frequency and high-frequency subband information on the performance of our MDPMFN model for infrared–visible remote sensing object detection, we conducted ablation experiments by removing either the high-frequency or the low-frequency fusion branch. As shown in Table V, when only low-frequency subbands were used (MDPMFN(no_HF)), the mAP dropped to 78.73%, indicating the loss of fine details such as edges and texture contrasts that are crucial for detecting small objects in drone imagery. Conversely, using only high-frequency subbands (MDPMFN(no_LF)) achieved a mAP of 80.63%, highlighting the strength of high-frequency information in local feature extraction, but also its limitation in capturing global context. In contrast, the full MDPMFN model, which integrates both low- and high-frequency subbands, achieved the highest mAP of 81.26%, outperforming MDPMFN(no_HF) and MDPMFN(no_LF) by 3.23% and 0.63%, respectively. These results confirm that jointly leveraging low-frequency and high-frequency subband information effectively balance global scene understanding and local discriminability. This enables suppression of background clutter and enhances precise object boundary localization, which is essential for multisensor remote sensing tasks involving scale-varying objects.

V. CONCLUSION

In this study, we propose a MDPMFN. This method effectively addresses the challenge of poor robustness in remote sensing object detection under complex environments. Specifically, we first design a PMFN framework, which achieves environment-adaptive multimodal feature fusion and significantly improves the robustness of the detection method. Second, we design a PGN that integrates prior information such as grayscale histograms and image entropy with multimodal image features to perceive environmental conditions. This enables the precise regulation of collaboration among subfusion experts optimized for different environmental conditions within the MoE fusion network framework. In addition, we propose a frequency-decoupled feature fusion network as the subfusion expert in the MoE fusion network framework, enabling personalized fusion of multimodal features across different frequency subbands. To enhance detection efficiency, we further introduce a cross-scale feature interleaved fusion strategy in the neck network responsible for cross-scale feature interaction. This strategy significantly reduces computational complexity while maintaining stable detection performance.

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Experimental results on the DroneVehicle and RGBT-Tiny datasets demonstrate that the proposed MDPMFN method achieves outstanding performance in remote sensing object detection tasks under complex environments.

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